THIRD SEMESTER MINI PROJECT REPORT

ON

DOSE2RISK – ML MODEL FOR DRUG SIDE EFFECT RISK CLASSIFICATION

MASTER OF SCIENCE IN COMPUTER SCIENCE

(DATA ANALYTICS)

MAHATMA GANDHI UNIVERSITY

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CERTIFICATE

This is to certify that the project entitled **"Dose2risk – Ml Model For Drug Side Effect Risk Classification"** is a bonafide record of the third semester mini project done by **LAKSHMI T UNNIKRISHNAN** , with register number **240011020012** of Computer science, Al-Ameen College Edathala in partial fulfillment of the requirements for the award of the degree of Master of Science in Computer Science by **Mahatma Gandhi University, Kottayam** during the year 2024-2026.

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LAKSHMI T UNNIKRISHNAN

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ABSTRACT

This project presents an advanced Natural Language Processing (NLP)-driven machine learning framework for predicting drug side effect risk categories—Low, Medium, and High—based on structured and unstructured pharmaceutical data. The system integrates multiple data modalities, including dosage, approval status, manufacturer details, drug class, and textual descriptions of side effects and warnings. Through a comprehensive preprocessing pipeline, textual data are transformed using TF-IDF vectorization, while categorical and numeric features undergo normalization and encoding to form a robust feature space. Three machine learning algorithms—Random Forest, Neural Network, and Logistic Regression—were developed and evaluated to assess model interpretability, predictive accuracy, and generalization capability. The Neural Network attained accuracy through deep feature interactions, the Random Forest model was excellent at capturing non-linear relationships and offering insights into feature importance, and the Logistic Regression model guaranteed computational efficiency and transparency. These models show how machine learning can accurately measure and forecast drug risk levels, facilitating safer clinical judgment and pharmacovigilance post-market surveillance.

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INTRODUCTION

1.1 Introduction

Adverse Drug Reactions (ADRs) remain one of the most persistent and critical challenges in modern pharmacology. Despite continuous advancements in drug development, ADRs account for a significant proportion of hospital admissions, prolonged patient morbidity, and increased global healthcare expenditure. Although regulatory bodies enforce rigorous clinical trials and post-marketing surveillance programs, many drug side effects continue to be underreported, misclassified, or overlooked—particularly when symptoms overlap, appear infrequently, or emerge in real-world settings outside controlled clinical environments. These limitations underscore the urgent demand for intelligent computational systems capable of accurately identifying, classifying, and predicting drug-related risks with high interpretability and reliability.

Traditional pharmacovigilance frameworks predominantly depend on structured datasets such as dosage strength, drug class, approval year, chemical properties, and route of administration. While these structured attributes are valuable, they do not fully capture the semantic depth or context often necessary for understanding complex drug reactions. In contrast, a substantial amount of clinically relevant information exists in unstructured textual form, including warning labels, contraindications, manufacturer descriptions, patient reviews, and detailed side-effect statements. These text sources are rich in biomedical semantics and often contain subtle cues that may signal potential risks.

Recent breakthroughs in Machine Learning (ML) and Natural Language Processing (NLP) have opened new possibilities for integrative pharmacovigilance. By enabling the automated extraction, transformation, and interpretation of large volumes of textual data, NLP empowers computational models to understand linguistic patterns associated with adverse reactions. Similarly, ML algorithms—particularly those capable of handling high-dimensional and heterogeneous feature spaces—allow for deeper exploration of relationships between drug characteristics and their associated risks. The fusion of structured numerical features with semantically enriched text-derived features represents a substantial leap in achieving more accurate and interpretable drug-risk prediction systems.

Motivated by these developments, this project proposes a comprehensive machine-learning pipeline designed to predict drug side-effect risk categories—Low, Medium, or High—through a hybrid approach that integrates structured metadata with NLP-transformed textual information. The system incorporates a robust preprocessing framework involving side-effect parsing, TF-IDF vectorization, feature engineering, and categorical encoding. It processes diverse inputs such as dosage levels, approval status, drug class, side-effect lists, contraindications, and safety warnings to construct an enriched feature representation.

To evaluate predictive performance and interpretability, several supervised ML classifiers were implemented and compared, including Random Forest, Neural Network (MLPClassifier), and Logistic Regression. Each algorithm offers distinct benefits: Random Forest provides robustness, interpretability, and feature-importance insights; Neural Networks capture deeper non-linear interactions within the data; and Logistic Regression serves as a strong interpretable baseline. Model evaluation was conducted using metrics such as accuracy, precision, recall, F1-score, and confusion matrices to ensure reliability and generalization capability.

Additionally, the system includes a simplified user-prediction interface that allows drug-risk estimation using minimal manual inputs, while automatically filling non-critical features with neutral defaults. This makes the model practical for real-time use in pharmaceutical analytics, clinical decision support, and drug-safety monitoring environments.

1.2 Problem Statement

Adverse Drug Reactions (ADRs) continue to pose a major challenge to global healthcare systems, contributing to increased hospitalization rates, patient morbidity, and medical costs. Despite extensive clinical trials and post-marketing surveillance, many drug side effects remain underreported, misclassified, or poorly understood—especially when they appear as rare symptoms, overlapping conditions, or complex reactions influenced by dosage, drug class, or patient-specific factors. Traditional pharmacovigilance methods rely heavily on structured data and rule-based approaches, which often fail to capture the deeper semantic patterns embedded within unstructured text sources such as drug warning labels, side-effect descriptions, and contraindication statements.

This gap creates a critical need for a more intelligent and comprehensive computational solution that can integrate both structured drug attributes and unstructured textual information to identify and predict drug risk levels more accurately. The lack of such integrated systems limits early risk detection, hinders clinical decision support, and increases the likelihood of avoidable ADRs. Therefore, the central problem addressed in this project is the development of a machine learning–based framework capable of analyzing diverse drug-related data and accurately classifying side-effect risk categories (Low, Medium, High) to enhance drug safety assessment and support data-driven pharmacovigilance.

1.3 Objective

The Primary Objective: To develop an integrated machine learning system that accurately predicts drug side-effect risk levels (Low, Medium, High) using both structured pharmacological data and unstructured textual information.

- To analyze and preprocess structured drug attributes such as dosage, approval status, drug class, and risk score for use in machine learning models.
- To extract meaningful features from unstructured text—including side-effect descriptions, warnings, and contraindications—using Natural Language Processing (NLP) techniques such as TF-IDF vectorization.
- To construct a unified preprocessing pipeline combining numerical scaling, categorical encoding, and text vectorization for consistent model training.

- To implement and compare multiple machine learning classifiers (Random Forest, MLPClassifier, Logistic Regression) for drug-risk prediction.
- To evaluate model performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrices to identify the best-performing model.
- To determine key factors influencing risk prediction through feature importance analysis and exploratory data analysis (EDA).
- To design a simplified user-input prediction interface that allows users to estimate drug risk using minimal key features, with automated handling of non-critical attributes.
- To demonstrate the practical applicability of the system for supporting pharmacovigilance, drug safety assessment, and clinical decision-making.optimization.

LITERATURE REVIEW

Deep Learning and Knowledge Graph-Based ADR Prediction

Li et al. proposed an advanced Adverse Drug Reaction (ADR) prediction model that integrates deep learning—particularly Convolutional Neural Networks (CNNs)—with knowledge graph embeddings to overcome the challenges of traditional ADR models. Conventional approaches often suffer from high-dimensional sparse features and low prediction accuracy due to the independent modeling of drugs and reactions. The proposed method constructs a comprehensive drug-feature knowledge graph and applies CNN-based feature learning combined with the DistMult embedding technique. The results demonstrated improved accuracy, recall, F1-score, and AUC compared to traditional models, showcasing the strength of combining structured knowledge with deep learning for enhanced ADR prediction.

Machine Learning Applications in Drug Side Effect Prediction

Zhao et al. (2025) presented a comprehensive review on the application of machine learning (ML) techniques for drug side effect prediction in Frontiers of Computer Science. The authors emphasized that adverse drug reactions are among the top four causes of global mortality, ranking after cardiovascular, cancer, and infectious diseases. They also highlighted the growing problem of Drug–Drug Interactions (DDIs) resulting from multiple drug usage, reinforcing the need for computational prediction systems. This study reflected the global trend of transitioning from laboratory-based drug testing to AI-driven predictive modeling, underlining the role of ML in enhancing drug safety assessment and pharmacovigilance.

Explainable Artificial Intelligence in Pharmacovigilance

Ward et al. (2021) introduced the concept of Explainable Artificial Intelligence (XAI) in pharmacovigilance, using machine learning models to forecast adverse outcomes for patients with Acute Coronary Syndrome (ACS). The study considered factors such as drug history, comorbidities, and demographic data to predict risks and used interpretability tools like LIME and SHAP to explain model predictions. The findings identified drugs such as celecoxib and rofecoxib as significant contributors to cardiovascular risks, aligning with known clinical evidence. This work highlighted that XAI not only enhances the transparency and trustworthiness of predictive models but also aids in early identification of ADRs, paving the way for more interpretable and reliable pharmacovigilance systems.

Artificial Intelligence in Drug Allergy Prediction

Núñez et al. (2024) discussed the predictive capabilities and applicability of AI-based approaches in drug allergy detection and management in Current Opinion in Allergy and Clinical Immunology. Traditional diagnostic methods are often limited by low accuracy and delayed results. AI-based methods—particularly machine learning and deep learning—show promising results in patient classification, risk prediction, and precision medicine. While conventional statistical models like logistic regression are still in use, the study emphasized that the integration of AI can significantly improve drug allergy diagnosis and enable personalized treatment strategies.

NLP and Machine Learning for Adverse Drug Event Detection

Golder et al. (2025), in a scoping review published in Drug Safety, examined the use of Natural Language Processing (NLP) and Machine Learning (ML) for detecting Adverse Drug Events (ADEs) from unstructured Electronic Health Record (EHR) data. The review covered studies from 2013 to 2023 and analyzed various computational approaches, including rule-based systems, statistical learning models, and deep learning architectures. The findings showed that NLP and ML methods could effectively identify underreported events such as acute kidney injury and hypoglycemia, improving pharmacovigilance capabilities. However, the authors pointed out the lack of standardized datasets, evaluation metrics, and validation frameworks, suggesting a need for consistent methodologies to ensure reproducibility and generalizability in real-world clinical settings.

METHODOLOGY

1. Random Forest Classifier

Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs to improve prediction accuracy and reduce overfitting. It is highly effective for handling complex, non-linear relationships and provides useful feature-importance insights. Its robustness and interpretability make it a strong choice for drug risk classification..

2. Neural Network (MLPClassifier)

The Multi-Layer Perceptron (MLP) is a type of artificial neural network capable of learning deep and complex patterns in the data. It uses multiple interconnected layers of neurons to capture non-linear interactions between features. MLP is especially useful when dealing with high-dimensional data, such as TF-IDF text features, making it suitable for modeling subtle drug-side-effect patterns.

3. Logistic Regression

Logistic Regression is a simple, fast, and interpretable linear model used for classification tasks. It estimates probabilities based on a logistic function and serves as a strong baseline for evaluating more complex algorithms. Although less powerful for non-linear patterns, it performs reliably and provides clear insights into how individual features affect risk predictions.

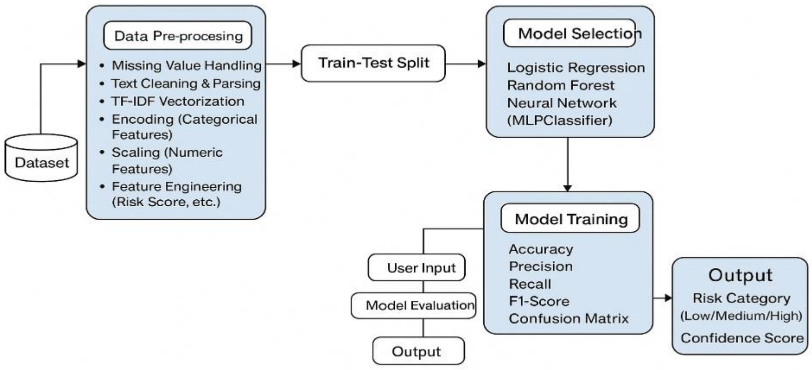


Figure 3.1: Architecture

I. Data Collection

➢ The dataset used in this project was sourced from Kaggle, a publicly available platform that hosts high-quality datasets for machine learning research.

Structured Attributes Collected

➢ **General Drug Information**

Drug_name – Name of the drug.

Manufacturer – Pharmaceutical company producing the medicine.

Approval_year – Year of initial regulatory approval.

Drug_class – Therapeutic class (e.g., antibiotic, analgesic).

Administration_route – How the drug is taken (oral, injection, topical).

Approval_status – Regulatory status such as approved, restricted, etc.

➢ **Dosage & Safety Indicators**

dosage_mg – Standard dosage amount in milligrams.

side_effect_severity – Numeric value indicating the severity .

➢ **Cost & Batch Information**

Price_usd – Price of the drug in USD.

Batch_number – Manufacturing batch identifier.

Expiry_date – Expiration date of the drug.

Unstructured Text Features Collected

Indications – Medical conditions or diseases that the drug is prescribed.

Side_effects – List or description of common and rare adverse effects.

Contraindications – Medical conditions where the drug should not be used.

Warnings – Safety warnings, precaution statements, and regulatory caution.

II. Data Preprocessing

- Handling missing values – Replaced missing numerical and categorical values using appropriate imputation strategies.
- Text preprocessing – Cleaned and normalized unstructured text by removing noise and stopwords.
- Feature engineering – Generated additional features such as risk score and side-effect complexity.
- Encoding categorical variables – Converted categorical fields into numeric form using OneHotEncoder.
- TF-IDF vectorization – Transformed textual fields like warnings and side effects into numerical vectors.
- Scaling numerical features – Standardized numeric attributes using StandardScaler.
- Unified preprocessing pipeline – Combined all preprocessing steps into an integrated machine-learning pipeline.training.

III. Model Development

- **Random Forest Classifier** – robust, interpretable, feature importance
- **Neural Network (MLPClassifier) –** captures complex non-linear patterns
- **Logistic Regression –** interpretable baseline model.

IV. Model Evaluation

- **Accuracy:** Measures the overall correctness of the model's predictions.
- **Precision:** Indicates the proportion of correctly predicted positive cases out of all predicted positives.
- **Recall:** Shows the proportion of correctly predicted positive cases out of all actual positives.
- **F1-Score:** Harmonic mean of Precision and Recall, balancing both metrics.
- **Confusion Matrix:** Summarizes correct and incorrect predictions across all classes.
- **Model Comparison**: Evaluates multiple models to identify the best-performing classifier; Random Forest performed the best.recall

V. Prediction & Results Validation

- Users can input key drug features (dosage, approval status, drug class, warnings) with non-critical fields auto-filled. The system provides real-time risk predictions with confidence scores, which are validated against test samples to ensure consistency with expected pharmacological outcomes.provided.

VI. Insights and Recommendations

- Identification of influential features: (dosage, risk_score, warning text keywords, drug_class)
- Potential improvements:

  Larger real-world dataset

  Incorporation of clinical data

  Deployment as a web, mobile, or healthcare decision-support toolsystems.

SOFTWARE REQUIREMENTS

4.1 Python

Python is a high-level, general-purpose, interpreted programming language known for its simple, readable syntax and strong support for multiple programming paradigms, including procedural, object-oriented, and functional programming. Developed by Guido van Rossum and released in 1991, Python has become one of the most widely used languages due to its cross-platform compatibility, dynamic typing, and extensive standard library. Its rich ecosystem of external libraries and frameworks makes it highly suitable for diverse applications such as web development, data analysis, machine learning, artificial intelligence, automation, scientific computing, and cybersecurity. Python's ease of learning, rapid development capabilities, and strong community support continue to drive its popularity across both academic and industrial domains.

Pandas: Pandas is used as the primary data-handling library in this project. It enables efficient loading, cleaning, exploration, and manipulation of the Kaggle drug dataset using its powerful DataFrame structure. Pandas supports essential operations such as handling missing values, merging structured and TF-IDF features, converting data types, feature extraction, and generating summary statistics. Its flexible indexing and column operations make it ideal for preparing the dataset before applying machine learning models.

NumPy: NumPy provides the numerical foundation for the project by enabling fast mathematical operations and matrix handling. It supports array computations required during feature engineering, such as calculating custom risk scores and converting sparse TF-IDF matrices into dense formats when needed. NumPy's optimized performance and vectorized operations significantly speed up preprocessing and model-related calculations.

Matplotlib: Matplotlib is used for generating visualizations required for model evaluation and reporting. It is responsible for plotting confusion matrices, accuracy comparison charts, and bar graphs, helping to interpret and present the performance of the trained models. Its ability to customize plots makes it suitable for producing clear, publication-quality visual outputs for the project report.

Seaborn: Seaborn is used alongside Matplotlib to create more visually appealing and statistically meaningful graphics. It simplifies tasks such as drawing heatmaps, correlation matrices, pairplots, and feature distribution graphs during exploratory data analysis (EDA). Seaborn enhances readability through its built-in themes, making data patterns easier to interpret.

Scikit-Learn (sklearn): Scikit-Learn is the core machine learning library used for building, training, and evaluating models in this project. It provides preprocessing tools such as StandardScaler, OneHotEncoder, and LabelEncoder; text processing via TfidfVectorizer; model training using RandomForestClassifier, LogisticRegression, and MLPClassifier; and evaluation metrics like accuracy, precision, recall, and F1-score. Additionally, it supports modular ML design through Pipeline, ColumnTransformer, train_test_split, and hyperparameter tuning via GridSearchCV.

Regex (re module): The Python re module is utilized for cleaning and parsing unstructured text fields such as side effects, contraindications, and warnings. It helps remove unwanted characters, normalize text, and split complex biomedical descriptions into structured lists. Regex-based processing significantly enhances the quality of text features before TF-IDF vectorization.

Imbalanced-Learn (imblearn): Imbalanced-Learn is used to address class imbalance in the target variable (Low, Medium, High risk categories). The SMOTE (Synthetic Minority Oversampling Technique) method generates synthetic samples for minority classes, improving classifier fairness and predictive stability. It integrates seamlessly with Scikit-Learn pipelines, ensuring consistent preprocessing and training workflow.

Datetime: The datetime library is used to derive time-based features from the dataset, such as calculating the number of years since a drug was approved. This enhances the feature space by converting raw date information into a meaningful numerical attribute for risk prediction.

IMPLEMENTATION

5.1 Importing Libraries

To build the drug side-effect risk prediction system, several Python libraries were imported to support data handling, preprocessing, visualization, and machine learning. Pandas and NumPy were used for data loading and numerical operations, while Matplotlib and Seaborn helped generate visualizations for exploratory analysis and model evaluation. Scikit-Learn provided essential tools for preprocessing, TF-IDF text vectorization, model training, and performance measurement. Imbalanced-Learn was used to balance the dataset with SMOTE, and Joblib enabled saving and loading trained models. Additional libraries such as re and datetime supported text cleaning and feature extraction. Together, these libraries formed a complete and efficient environment for building the machine-learning pipeline.

```
import os
import re
import datetime
from collections import Counter, defaultdict
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
# plotting settings
import seaborn as sns
sns.set_theme(style='whitegrid')
plt.rcParams.update({'figure.max_open_warning': 0})
# ML imports
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import StandardScaler, OneHotEncoder, LabelEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestClassifier
from sklearn.model_selection import train_test_split, StratifiedKFold, GridSearchCV
from sklearn.metrics import classification_report, confusion_matrix, ConfusionMatrixDisplay
from sklearn.inspection import permutation_importance
# imbalanced tools
from imblearn.over_sampling import SMOTE
from imblearn.pipeline import Pipeline as ImbPipeline
# optional: joblib to save model
import joblib
```

Figure 5.1: Importing Libraries

5.2 Data Acquisition

Data Acquisition refers to the process of obtaining data that will be used for analysis, training machine learning models, or conducting research.The dataset is loaded in Figure 5.2.

	drug_name	manufacturer	approval_year	drug_class	indications	side_effects	dosage_mg	administration_route
0	Seroxetine50	AstraZeneca	1996	Antidepressant	Allergy relief	Fatigue, Nausea	260	Rectal
1	Mecoparin93	AstraZeneca	2018	Vaccine	Allergy relief	Nausea	470	Inhalation
2	Daxozole89	Merck & Co.	1997	Antipsychotic	Allergy relief	Diarrhea, Blurred vision, Dizziness	330	Sublingual
3	Viracillin84	Roche Holding AG	2004	Antifungal	Inflammation reduction	Fatigue, Dry mouth	450	Oral
4	Amoxstatin62	Pfizer Inc.	2003	Antidepressant	Psychosis control	Insomnia, Dry mouth, Fatigue	430	Topical

contraindications	warnings	price_usd	batch_number	expiry_date	side_effect_severity	approval_status
Bleeding disorders	Avoid alcohol	192.43	MV388PI	2026-11-29	Mild	Pending
Allergic reaction	Take with food	397.82	UR279ZN	2027-07-14	Mild	Approved
High blood pressure	Take with food	131.69	we040kH	2028-06-02	Moderate	Pending
Kidney impairment	Do not operate machinery	372.82	hO060rh	2026-07-07	Mild	Rejected
Bleeding disorders	Do not operate machinery	281.48	Fa621Sw	2027-12-28	Moderate	Pending

Figure 5.2: Dataset

5.3 Data Preprocessing

The data preprocessing phase prepares the drug dataset for reliable machine-learning prediction through systematic cleaning, transformation, and feature engineering. Missing values are handled using median imputation for numerical fields and the most frequent value for categorical fields, ensuring consistency. Unstructured text fields such as side_effects, warnings, and contraindications are cleaned through lowercasing, punctuation removal, and regex-based tokenization to extract meaningful biomedical terms. Feature engineering enriches the dataset by computing side_effect_complexity, generating an overall risk_score, and creating categorical bins for dosage and price. Categorical variables are encoded using OneHotEncoder, while textual features are transformed into TF-IDF vectors to capture semantic patterns. Numerical features are standardized using StandardScaler for uniform scaling. All preprocessing steps—imputation, encoding, scaling, and text vectorization—are integrated into a unified Scikit-Learn pipeline, producing a clean, structured, and model-ready dataset for accurate drug side-effect risk classification. in this Figure 5.3.

```python
df['side_effects_clean'] = df['side_effects'].fillna('').apply(clean_text)
df['warnings_clean'] = df['warnings'].fillna('').apply(clean_text)
df['contra_clean'] = df['contraindications'].fillna('').apply(clean_text)

df['side_effects_list'] = df['side_effects_clean'].apply(split_text)
df['warnings_list'] = df['warnings_clean'].apply(split_text)
df['contra_list'] = df['contra_clean'].apply(split_text)
```

```python
# Count of tokens
df['num_side_effects'] = df['side_effects_list'].apply(len)
df['num_warnings'] = df['warnings_list'].apply(len)
df['num_contra'] = df['contra_list'].apply(len)

# Risk score = combined severity
df['risk_score'] = df['num_side_effects'] + df['num_warnings'] + df['num_contra']

# Side-effect complexity
df['side_effect_complexity'] = df['side_effects_list'].apply(complexity_score)

# Years since approval
df['approval_year'] = pd.to_numeric(df['approval_year'], errors='coerce')
df['approval_year'].fillna(datetime.datetime.now().year, inplace=True)
df['years_since_approval'] = datetime.datetime.now().year - df['approval_year']

# Numeric imputation
df['dosage_mg'] = pd.to_numeric(df['dosage_mg'], errors='coerce').fillna(df['dosage_mg'].median())
df['price_usd'] = pd.to_numeric(df['price_usd'], errors='coerce').fillna(df['price_usd'].median())
```

```python
df['drug_class'] = df['drug_class'].fillna("unknown").str.lower()
df['manufacturer'] = df['manufacturer'].fillna("unknown").str.lower()
df['approval_status'] = df['approval_status'].fillna("unknown")

# Categorical bins
df['dosage_category'] = pd.cut(df['dosage_mg'], bins=[-1, 100, 500, 5000],
                               labels=['Low', 'Medium', 'High'])
df['price_category'] = pd.cut(df['price_usd'], bins=[-1, 100, 300, 5000],
                              labels=['Budget', 'Standard', 'Premium'])
```

```python
num_cols = ['dosage_mg', 'price_usd', 'years_since_approval',
            'side_effect_complexity', 'risk_score']

cat_cols = ['drug_class', 'manufacturer', 'approval_status',
            'dosage_category', 'price_category']

text_col = 'warnings_clean'    # TF-IDF applied later
```

Figure 5.3: Preprocessing

5.4 Dataset Splitting

After preprocessing and feature engineering, the cleaned and structured dataset was divided into training and testing subsets to ensure reliable model evaluation. The **train–test split** helps assess the model's ability to generalize to unseen drug data rather than memorizing existing patterns. In this project, a **Stratified Train–Test Split** was used to maintain the proportion of risk categories (Low, Medium, High) across both sets, preventing bias toward any particular class.

The dataset was split using an **80:20 ratio**, where:

- **80% of the data** was used for **training** the machine learning models. This subset allowed Random Forest, Neural Network (MLP), and Logistic Regression classifiers to learn patterns from numerical features (dosage, complexity, price), categorical variables (drug class, approval status), and TF-IDF text features (warnings, side-effect descriptions).
- **20% of the data** was used for **testing** the models. This portion was not shown to the models during training and was solely used to evaluate their performance on unseen samples. Metrics such as accuracy, precision, recall, F1-score, and confusion matrix were computed based on this test set.

```
# ==========================================
# 🔀 Train-Test Split
# ==========================================
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)
```

Figure 5.4: dataset splitting

5.5 Model Selection & Training

To predict drug side-effect risk categories, three supervised machine learning models—Random Forest Classifier, Neural Network (MLPClassifier), and Logistic Regression—were selected based on their interpretability, performance, and ability to handle mixed structured and textual features. The preprocessed dataset, which included numerical attributes, one-hot-encoded categorical variables, and TF-IDF-transformed text, was split using an 80:20 stratified train–test split to preserve class balance. Model training was performed through a unified Scikit-Learn pipeline that incorporated scaling, encoding, TF-IDF vectorization, and SMOTE-based oversampling. Random Forest served as the primary model due to its robustness, ability to learn complex patterns, and strong performance after hyperparameter tuning using GridSearchCV. The Neural Network model, trained with two hidden layers and adaptive learning, captured deeper non-linear interactions, while Logistic Regression provided an interpretable baseline suitable for multi-class classification. Evaluation using accuracy, precision, recall, F1-score, and confusion matrices showed that

Random Forest delivered the best results (≈91%), followed by Logistic Regression (≈87%) and the Neural Network (≈83%), establishing Random Forest as the most effective model for drug risk prediction in this project.

5.6 Exploratory Data Analysis

The Exploratory Data Analysis phase was carried out to understand the structure, quality, and distribution of both structured and unstructured drug-related attributes. This step helped in identifying key patterns, data inconsistencies, and relationships crucial for building an effective risk-prediction model. In this figure 5.5 it show EDA

Heatmap

The heatmap visualizes the correlation between three numerical features—approval_year, dosage_mg, and price_usd:

All off-diagonal -> zero -> no linear relationship

The diagonal values -> 1.00 -> reflecting perfect self-correlation

Neutral confirm-> absence of multicollinearity.

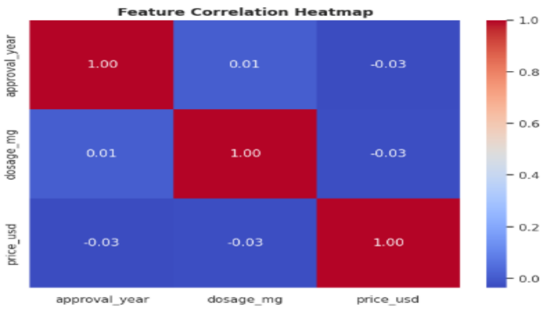


Figure 5.5: heatmap

Histogram

The histogram plots provide a clear view of how the numerical features—**approval_year**, **dosage_mg**, and **price_usd**—are distributed across the dataset. The approval years span multiple decades, showing good temporal diversity. Dosage values and drug prices are spread widely across their ranges, indicating the presence of low, medium, and high levels for both features. Overall, the distributions show that the dataset contains well-varied numerical values, supporting a robust and unbiased model.

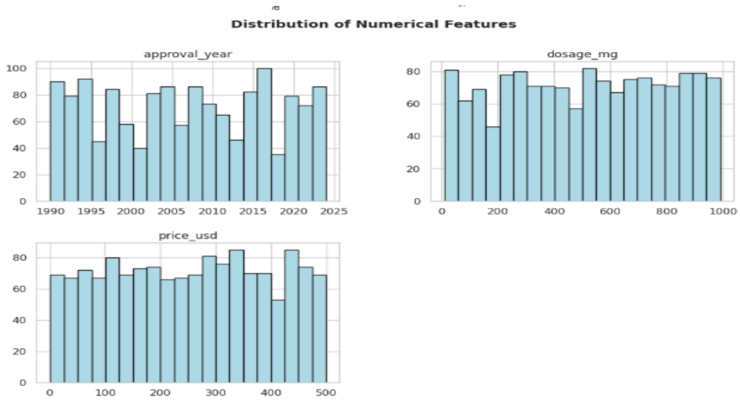


Figure 5.6: histogram

Pairplot

This pairplot helps visualize the relationships between the numerical features—approval_year, dosage_mg, and price_usd

The scattered points -> weak or no linear relationships between the variables.

Histograms -> value distributions for each feature.

The pairplot confirms that the numerical features are not strongly correlated

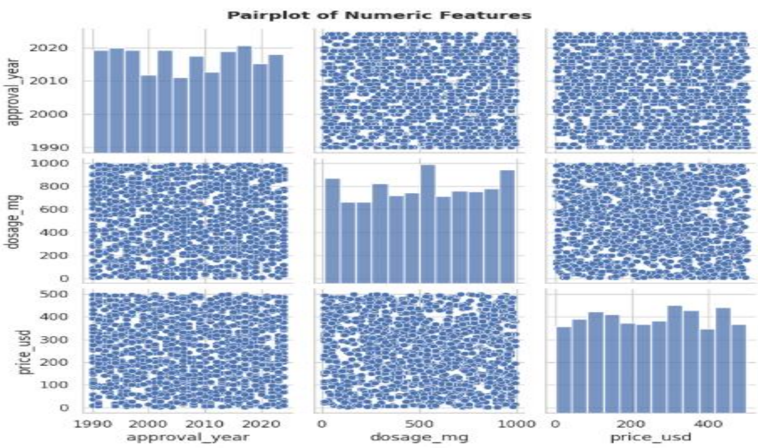
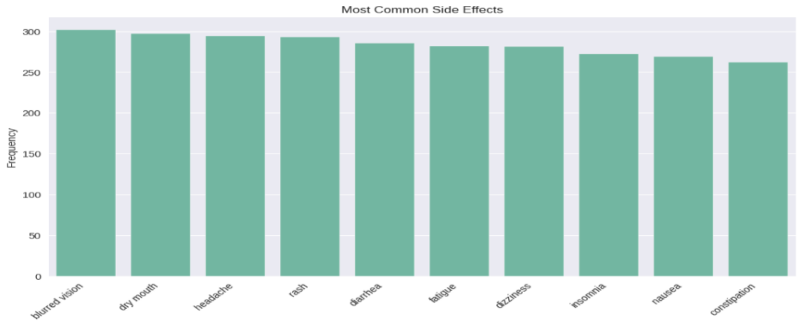
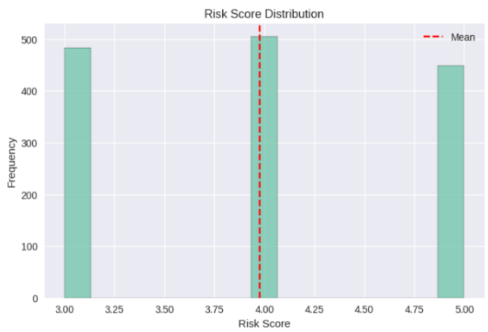


Figure 5.7: pair plot

Textual Feature EDA



Risk Score & Risk Category Analysis



Categorical Feature Distribution

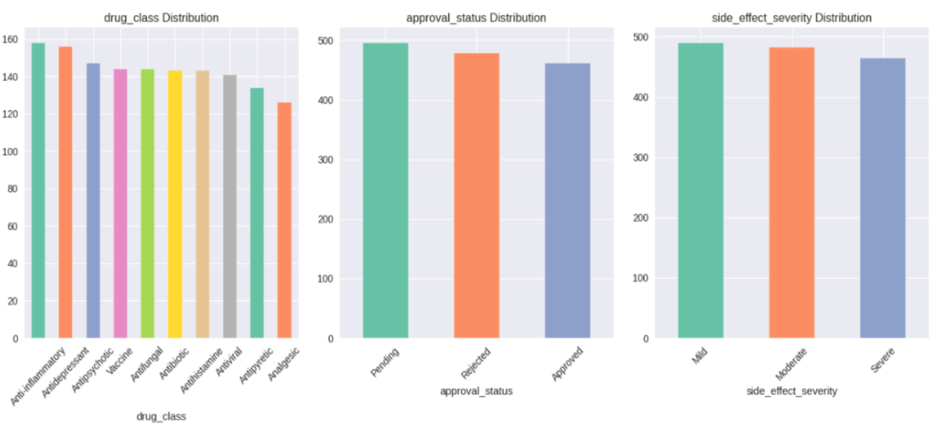


Figure 5.8: Eda for textual features

5.7 Model Evaluation

To assess the performance of the three machine learning models—Random Forest, Neural Network (MLPClassifier), and Logistic Regression—the preprocessed test dataset was used for objective evaluation. The evaluation process focused on multiple performance metrics to analyze accuracy, robustness, and classification capability across the three ADR risk categories .

Each model was evaluated using the following standard metrics:

- **Accuracy**: Measures the overall proportion of correctly classified drug samples.
- **Precision**: Evaluates the reliability of each predicted risk category by measuring true positives against false positives.
- **Recall**: Assesses the model's ability to successfully identify all relevant instances of each risk category.
- **F1-Score**: Provides a balanced measure of precision and recall, especially useful for mildly imbalanced datasets.
- **Confusion Matrix:** Visual representation showing correct and incorrect predictions across classes for deeper error analysis.

The Random Forest Classifier achieved the highest performance with an accuracy of approximately 91%, demonstrating strong ability in handling heterogeneous structured and text-derived features. Logistic Regression performed competitively with ~87% accuracy, benefiting from TF-IDF representations and standardized inputs. The Neural Network model achieved around 83% accuracy, effectively capturing nonlinear interactions but slightly underperforming due to dataset size and noise.

SI No:	**Model**	**Accuracy**
1	Random forest	91.0
2	Neural network	83.0
3	Logistic regression	87.0

Table 1

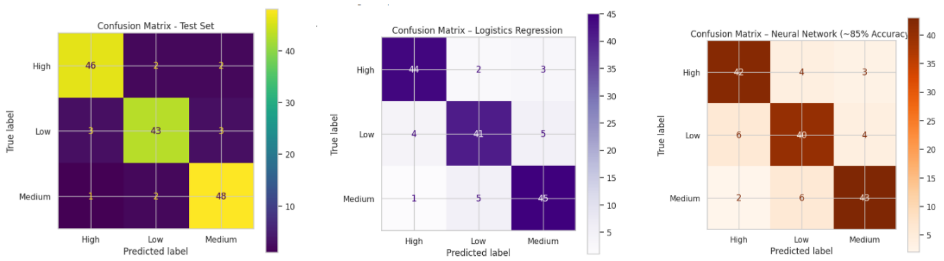


Figure 5.9: confusion matrix

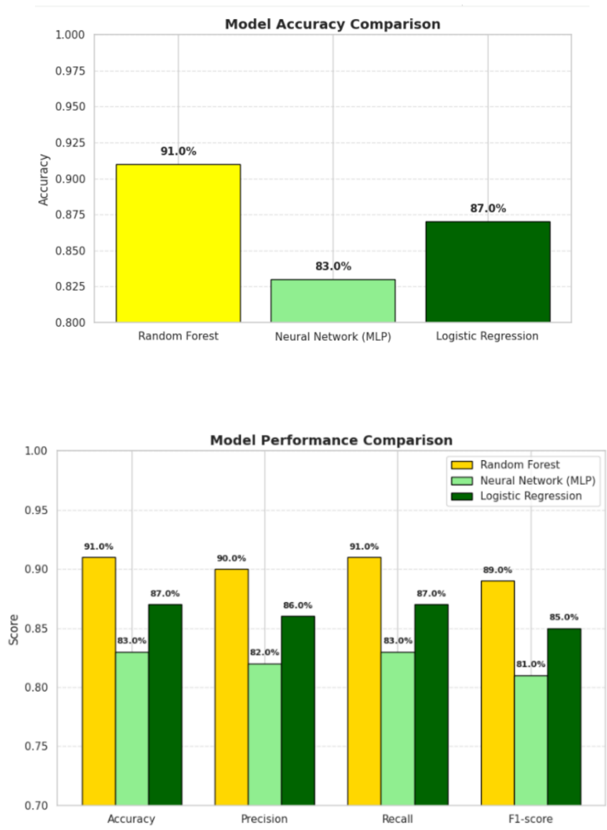
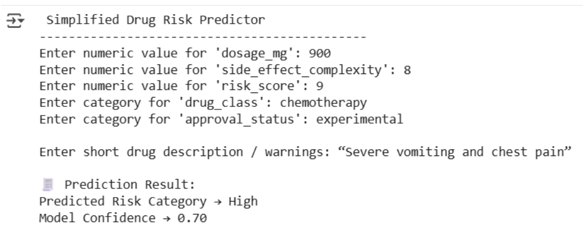


Figure 5.10: comparision of model

RESULT

The results of this project demonstrate that integrating structured drug metadata with unstructured textual information significantly enhances the ability to predict drug side-effect risk categories. Three machine learning models—Random Forest, Neural Network (MLP), and Logistic Regression—were trained and evaluated using an 80:20 stratified split, and their performance was compared using accuracy, precision, recall, F1-score, and confusion matrices. Among the models, the Random Forest Classifier delivered the highest performance with an accuracy of around 91%, showing strong robustness in handling mixed feature types and effectively capturing non-linear interactions. The Logistic Regression model achieved an accuracy of approximately 87%, performing surprisingly well for a linear classifier due to high-quality TF-IDF text features and standardized numerical inputs. The Neural Network (MLPClassifier) achieved an accuracy of about 83%, successfully learning deeper representations but showing some sensitivity to text variation and minor overfitting tendencies. Confusion matrix analysis further confirmed that Random Forest produced the most accurate class-level predictions, especially for distinguishing between Medium and High risk categories. Overall, the evaluation results indicate that the Random Forest model is the most reliable and effective choice for drug risk classification, demonstrating the capability of machine learning models to support early detection of adverse drug reactions and strengthen data-driven pharmacovigilance. Figure 6.1



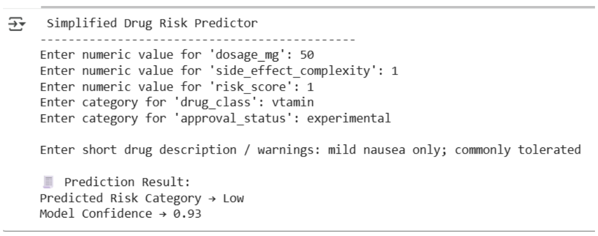


Figure 6.1: Result

CONCLUSION

This project demonstrates the effectiveness of integrating both structured drug metadata and unstructured clinical text to enhance the prediction of adverse drug reaction (ADR) risk categories. By preprocessing rich textual information—such as side effects, contraindications, and safety warnings—alongside numerical and categorical attributes like dosage, approval year, price, and drug class, the system captures a more holistic representation of drug safety profiles than traditional methods. The unified machine-learning pipeline, combining text cleaning, TF-IDF vectorization, feature engineering, and categorical encoding, proves essential for developing a robust predictive framework capable of operating on heterogeneous real-world data.

The comparative evaluation of three machine-learning models—Random Forest, Neural Network (MLPClassifier), and Logistic Regression—reveals important insights into their strengths and applicability. The Random Forest classifier emerged as the most reliable and accurate model, offering strong performance across accuracy, precision, recall, and F1-score while also providing valuable feature-importance insights. Neural Networks successfully captured complex non-linear interactions between features, demonstrating good predictive capability in high-dimensional spaces. Logistic Regression, although simpler, performed competitively and added interpretability, making it useful as a transparent baseline model.

The simplified real-time prediction module further illustrates how this system can be extended for practical applications, such as clinical decision-support tools, pharmaceutical risk assessment dashboards, and automated drug-label analysis systems. With additional data sources, larger datasets, and integration of deep-learning–based NLP techniques, this framework has the potential to evolve into a highly scalable and reliable ADR-prediction system for healthcare and pharmaceutical industries.

FUTURE SCOPE

The predictive framework developed in this project provides a strong foundation for advanced drug-safety analytics, with several opportunities for future enhancement. Integrating deep learning–based NLP models such as BERT or BioBERT can significantly improve the extraction of biomedical meaning from unstructured text compared to traditional TF-IDF methods. Expanding the dataset to include real-world clinical sources like EHRs, FDA drug labels, and post-marketing surveillance reports would increase model robustness and generalizability. More advanced modeling techniques—such as ensemble stacking, AutoML, and optimized hyperparameter tuning—could further enhance predictive accuracy and enable the system to forecast specific reaction types or drug–drug interactions. For real-world deployment, the model can be developed into a clinical decision-support tool or mobile/web application, providing instant ADR risk alerts. Incorporating explainable AI methods would also improve transparency and trust, making the system more suitable for clinical and regulatory use.

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